

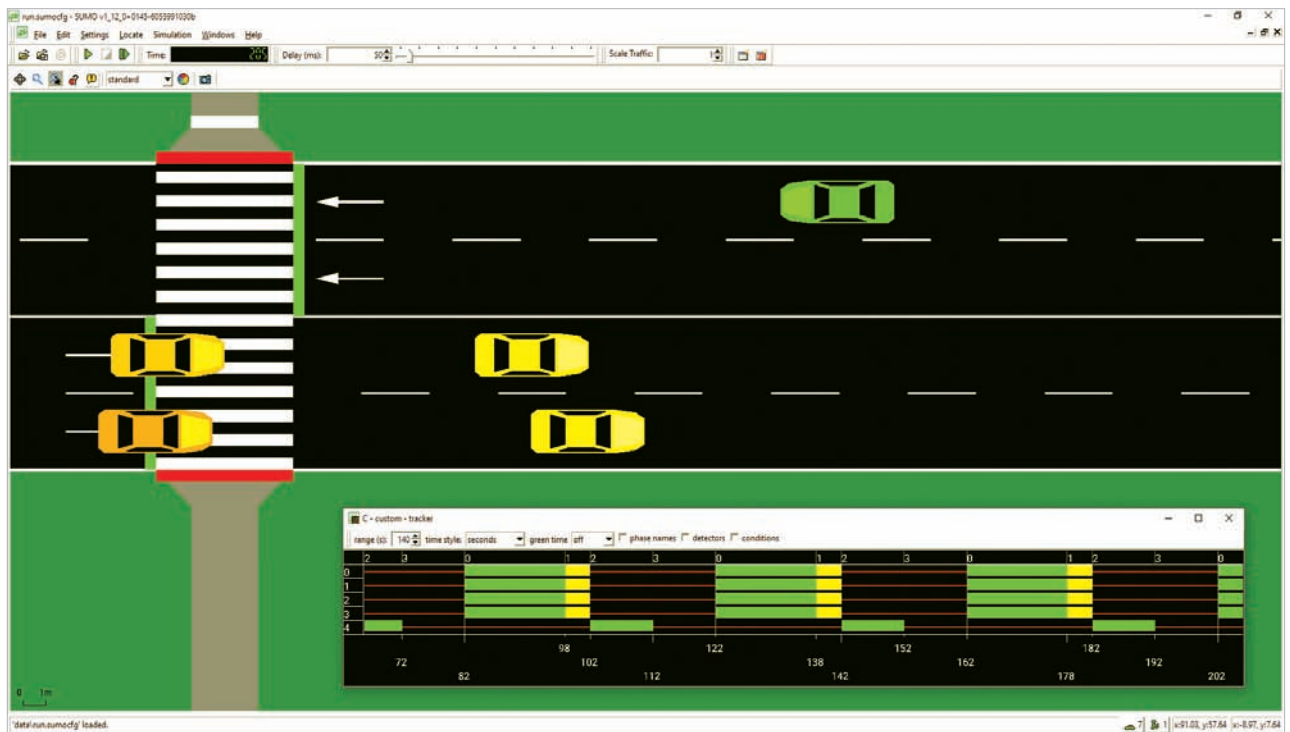


Figure 5: Smart vehicular simulation in SUMO

calculation, are included with SUMO and automate fundamental processes for the development, execution, and evaluation of traffic simulations. Custom models can be added to SUMO, and it offers a number of APIs for controlling the simulation from a distance.

The key features of SUMO are:

- Free and open source
- Automated driving
- Vehicle communication
- Traffic management
- Multimodal traffic
- Microscopic simulation
- Online interactions
- Traffic lights

- Network import
- Demand generation
- Performance
- Portability

The manufacturing sector is experiencing exciting times right now as new digital technologies emerge. Industry 4.0 is also making it easier for humans and robots to collaborate, enabling firms to gain more knowledge, lower the possibility of error, and

make wiser decisions. The only way for manufacturers to stay ahead of the curve in this new digital era is by embracing Industry 4.0.

Researchers and practitioners of Industry 4.0 can use free and open source platforms to simulate and experiment with its real world applications. Such simulation sessions will help analyse the outcome in real-time before actual deployment. **END** 

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